# When the three good events hold, the integer-friendly random integer coincides with Arratia's: J_x = J_x^*

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*Editorial note.* The counted statement is the author's labelled lemma 'lem:decomposition J neq J*' (source lines 842-848); the proof environment begins on the next line after \end{lem} apart from one blank line, so the pairing is by adjacency.

*Editorial note.* The retained context is the opening of the author's own section on the integer-friendly version of J_x, which defines the von Mangoldt weight h that both the statement and the proof use.

*Editorial note.* No counter adjustment is retained: the publisher numbering is not reproduced and the item prints with the wrapper's own sequential numbering.

*Editorial note.* No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

*Editorial scope.* One original result of the pinned publisher article, counted once: the event decomposition forcing J_x = J_x^* (author label lem:decomposition J neq J*).

*Allowed standard prerequisites (editorial).*

- The two random integers. J_x = prod_{i>=1} e^{h(Y_i)} is the integer-friendly product, and J_x^* is Arratia's random integer defined by the ordered indexing of the transformed point process R^*.. The statement compares the two and the proof computes both products over the index sets B_+ and B_-.
- The good events E_1, E_2, E_3. E_1, E_2, E_3 are the events displayed immediately before the lemma, controlling the spacing sums S_1, S_0, the error R_x and the ordering of the ratios W_i Y_i / h(Y_i).. Each event is consumed at a different place: E_1 at lines 850-864, E_2 and E_3 at lines 866-877.
- von Mangoldt function. Lambda(n) = log p if n is a prime power p^k and 0 otherwise.. It is the weight h used in the products J_x and in the section's construction, retained in the context.

# 1 An integer-friendly version of $J_x$

Let $\Lambda$ be the von Mangoldt function, that is to say

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \text{ for some prime power } p^k, \\ 0 & \text{otherwise.} \end{cases}$$

**Lemma 1.1.** *For $x > 1$, we have $\mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3 \subseteq \{J_x = J_x^*\}$. In particular,*

$$\mathbb{P}\big[J_x \neq J_x^*\big] \leq \mathbb{P}[\mathcal{E}_1^c] + \mathbb{P}[\mathcal{E}_1 \cap \mathcal{E}_2^c] + \mathbb{P}[\mathcal{E}_1 \cap \mathcal{E}_3^c].$$

*Proof.* Recall that $J_x = \prod_{i\geq 1} e^{h(Y_i)}$ and assume that $\mathcal{E}_1$ occurs. Then

$$\log J_x = \sum_{i=1}^{\infty} h(Y_i) \leq \sum_{i\geq 1} \big(Y_i + r(Y_i)\big) \leq S_1 + R_x + r_0 < \log x, \tag{1}$$

since $r(Y_1) \leq r_0$. Similarly, we have

$$\log\big(J_x \cdot e^{h(Y_0)}\big) = \sum_{i=0}^{\infty} h(Y_i) \geq \sum_{i\geq 0} \big(Y_i - r(Y_i)\big) \geq S_0 - R_x - 2r_0 > \log x.$$

Therefore, $\mathcal{E}_1$ implies the inequalities

$$J_x < x < J_x e^{h(Y_0)}. \tag{2}$$

In particular, we must have $Y_0 > e^{-\gamma}$.

Assume now further that $\mathcal{E}_2$ and $\mathcal{E}_3$ also occur. We claim that this implies $J_x = J_x^*$. Let $B_+$ be the set of integers $i \geq 1$ such that $Y_i > e^{-\gamma}$, and let $B_-$ be the set of integers $i \leq -1$ such that $Y_i > e^{-\gamma}$. By our assumption that $\mathcal{E}_2 \cap \mathcal{E}_3$ occurs and by the definition of $\eta$, we have

$$\frac{W_i Y_i}{h(Y_i)} > \frac{W_0 Y_0}{h(Y_0)} \qquad \text{if } i \in B_+ \tag{3}$$

and, similarly,

$$\frac{W_i Y_i}{h(Y_i)} < \frac{W_0 Y_0}{h(Y_0)} \qquad \text{if } i \in B_-. \tag{4}$$

Now, let $K \geq 0$ and $(T_j^*, Q_j^*)$ with $j \in \mathbb{Z}_{\leq K}$ be the indexing of points of $\mathscr{R}^*$ given in Section 1, that is to say we have $T_{j-1}^* < T_j^*$ for all $j \in \mathbb{Z}_{\leq K}$, and such that $\prod_{j=0}^{K} Q_j^* > x \geq \prod_{j=1}^{K} Q_j^*$. Let $j_0 \in \mathbb{Z}_{\leq K}$ be such that $T_{j_0}^* = W_0 Y_0 / h(Y_0)$ (which exists because $Y_0 > e^{-\gamma}$). Using relations (3) and (4), we find

$$\prod_{j_0 < j \leq K} Q_j^* = \prod_{i \in B_+} e^{h(Y_i)} = \prod_{i \geq 1} e^{h(Y_i)} = J_x.$$

On the other hand, we have $\prod_{j_0 \leq j \leq K} Q_j^* = e^{h(Y_0)} J_x$. Hence, using (2), we find that $\prod_{j_0 < j \leq K} Q_j^* < x < \prod_{j_0 \leq j \leq K} Q_j^*$. In particular, $j_0 = 0$ and thus $J_x^* = \prod_{j_0 < j \leq K} Q_j^* = J_x$, as claimed. □

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